

Xiangyu Liu

CONTACT INFORMATION	111 8th Ave, https://xiangyu-liu.github.io/	xiangyuliuxyl@google.com
EMPLOYMENT	Google LLC, NY, USA <i>Research Scientist</i> Work in the area of AI and machine learning for the cutting-edge problems of Google	Starting from Aug. 2026
EDUCATION	University of Maryland, College Park, MD, USA <i>Ph.D. in Computer Sciences</i> • Advisor: Kaiqing Zhang Shanghai Jiao Tong University (SJTU), Shanghai, CN <i>Bachelor in Computer Science</i> • Zhiyuan Honors Program of Engineering (an elite program for top 5% talented students) University of California, Berkeley, CA, USA <i>Exchange student</i> • GPA: 4.0/4.0	Aug. 2021 – Jun. 2026 Sep. 2017 – Jun. 2021 Jan. 2020 – May 2020
RESEARCH INTERESTS	• Multi-agent RL and game theory: paper [J1, C1, C2, C6, C8] • LLM agents, alignment: paper [C7, C9, C10] • Robust RL: paper [C3, C4, C5]	
INTERN EXPERIENCES	Google Research, CA, USA <i>Research intern with Market Algorithm team</i> <i>Working with Zhe Feng, Aranyak Mehta, Di Wang</i> • Fundamental research on inference-time scaling techniques for LLM strategic decision-making, with an emphasis on the multi-agent negotiation games • Developed a best-of-N (BoN) sampling with opponent simulation method, enabling feedback-driven in-context self-improving agents in repeated interaction • Paper [C10] under submission Bloomberg AI Group, NY, USA <i>Research intern</i> • Research on bond pricing with recurrent neural networks • Developed a novel attention mechanism for dynamically capturing correlations among multiple time series	May. 2025 – Dec. 2025 June. 2022 – Sep. 2022

JOURNAL
PUBLICATIONS

- [J1] **Xiangyu Liu**, Kaiqing Zhang
Partially Observable Multi-Agent Reinforcement Learning with Information Sharing
SIAM Journal on Control and Optimization (SICON 2025)
Shorter version appeared at ICML 2023

CONFERENCE
PUBLICATIONS

- [C10] **Xiangyu Liu**, Di Wang, Zhe Feng, Aranyak Mehta
Scaling Inference-Time Computation Via Opponent Simulation: Enabling Online Strategic Adaptation for Repeated Negotiation
International Conference on Machine Learning (ICML 2026)
- [C9] (*denotes equal contribution) Chanwoo Park*, **Xiangyu Liu***, Asuman E. Ozdaglar, Kaiqing Zhang
Do LLM Agents Have Regret? A Case Study in Online Learning and Games
International Conference on Learning Representations (ICLR 2025)
Oral talk at ICLR 2024 workshop: How Far Are We From AGI?
- [C8] (Alphabetical order) **Xiangyu Liu**[†], Haoyi You[†], Kaiqing Zhang[†]
Principled Learning-to-Communicate with Quasi-Classical Information Structures
IEEE Conference on Decision and Control (CDC 2025)
- [C7] Pankayaraj Pathmanathan, Souradip Chakraborty, **Xiangyu Liu**, Yongyuan Liang, Furong Huang
Is Poisoning a Real Threat to LLM Alignment? Maybe More So Than You Think
AAAI Conference on Artificial Intelligence (AAAI 2025)
- [C6] (Alphabetical order) Yang Cai[†], **Xiangyu Liu**[†], Argyris Oikonomou[†], Kaiqing Zhang[†]
Provable Partially Observable Reinforcement Learning with Privileged Information
Conference on Neural Information Processing Systems (NeurIPS 2024)
- [C5] Yongyuan Liang, Yanchao Sun, Ruijie Zheng, **Xiangyu Liu**, Tuomas Sandholm, Furong Huang, Stephen McAleer
Game-theoretic Robust Reinforcement Learning Handles Temporally-coupled Perturbations
International Conference on Learning Representations (ICLR 2024)
- [C4] **Xiangyu Liu**, Chenghao Deng, Yanchao Sun, Yongyuan Liang, Furong Huang
Beyond Worst-case Attacks: Robust RL with Adaptive Defense via Non-dominated Policies
International Conference on Learning Representations (ICLR 2024) *Spotlight (top 5%)*
- [C3] **Xiangyu Liu**, Souradip Chakraborty, Yanchao Sun, Furong Huang
Rethinking Adversarial Policies: A Generalized Attack Formulation and Provable Defense in RL

International Conference on Learning Representations (**ICLR 2024**)

Outstanding Paper Award at NeurIPS 2022 Workshop on Trustworthy and Socially Responsible Machine Learning.

[C2] **Xiangyu Liu**, Kaiqing Zhang

Partially Observable Multi-agent RL with (Quasi-)Efficiency: The Blessing of Information Sharing

International Conference on Machine Learning (**ICML 2023**)

[C1] **Xiangyu Liu**, Hangtian Jia, Ying Wen, Yujing Hu, Yingfeng Chen, Changjie Fan, Zhipeng Hu, Yaodong Yang

Towards Unifying Behavioral and Response Diversity for Open-ended Learning in Zero-sum Games

Conference on Neural Information Processing Systems (**NeurIPS 2021**)

HONORS AND AWARDS

Outstanding Paper Award, NeurIPS 2022 Workshop on Trustworthy and Socially Responsible Machine Learning. 2022

Dean's Fellowship, University of Maryland, College Park. 2021

National Scholarship (Top 0.2% in China), Ministry of Education of P.R.China. 2018&2019

1st Prize in Chinese College Mathematics Competitions (Top 1 at SJTU, selected for final). 2018

A-class Scholarship for Excellent Academic Performance (Top 1% at SJTU), SJTU. 2018

TALKS

UVA RL meetup, online, 2025

Invited talk on partially observable RL with privileged information (paper [C6])

2024 INFORMS Optimization Society Conference (IOS 2024), Houston, Texas, 2024

Invited talk on multi-agent RL (paper [J1] and [C2])

TSRML workshop of NeurIPS 2022

Contributed talk on adversarial policies in competitive games (paper [C3])

RL China seminar series, online, China

Invited talk on unifying diversity in open-ended learning for zero-sum games (paper [C1])

ACADEMIC SERVICES

Conference reviewer for NeurIPS 2024-2025, ICML 2025, ICLR 2025, AISTATS 2025, UAI 2024-2025, AAMAS 2025, CDC 2025

TEACHING EXPERIENCE

Student organizer of summer AI camps at UMD for K-12 students Summer 2023/2024

TA for cryptotography Spring 2022

TA for common sense reasoning in NLP Fall 2021

COMPUTER SKILLS

- Programming Languages: Python, C/C++, Java, MATLAB, $\text{\LaTeX}$
- Deep Learning Packages: PyTorch, TensorFlow